

- (i) Bepaal die versnelling van die twee blokke.
Determine the acceleration of the two blocks.

$$\begin{aligned}
 m_1 \quad \Sigma F_x &= T - m_1 g \sin \theta = m_1 a & T &= m_1 a + m_1 g \sin \theta \\
 m_2 \quad \Sigma F_x &= T - m_2 g \sin \theta = -m_2 a & T &= m_2 g \sin \theta - m_2 a \\
 m_1 a + m_1 g \sin \theta &= m_2 g \sin \theta - m_2 a \\
 3,5 a + 3,5(9,8) \sin 35 &= 8(9,8) \sin 35 - 8 a \\
 11,5 a &= 64,64 \\
 a &= 5,62 \text{ m/s}^2
 \end{aligned}$$

(4)

- (ii) Bepaal die spankrag in die tou.
Determine the tension in the string.

$$\begin{aligned}
 T &= 3,5(5,62) + 3,5(9,8) \sin 35 \\
 &= 0 \text{ N}
 \end{aligned}$$

(2)

Wanneer daar wrywing tussen die blokke en die oppervlak is, is die versnelling $a = 1,5 \text{ m/s}^2$. Indien die wrywingskoeffisiënte dieselfde is vir beide blokke, bepaal:
With friction between the blocks and the surfaces the acceleration is $a = 1,5 \text{ m/s}^2$. If the coefficients of kinetic friction between the blocks and the inclines are the same, determine:

- (iii) Bepaal die kinetiese wrywingskoeffisiënt μ_k .
Determine the value of coefficient of kinetic friction μ_k .

$$\begin{aligned}
 f_k &= \mu_k N & \mu_k &= \frac{f_k}{N} = \frac{m g \sin \theta}{m g \cos \theta} \\
 & & &= \tan \theta = \tan 35 \\
 \mu_k &= 0,7
 \end{aligned}$$

(6)